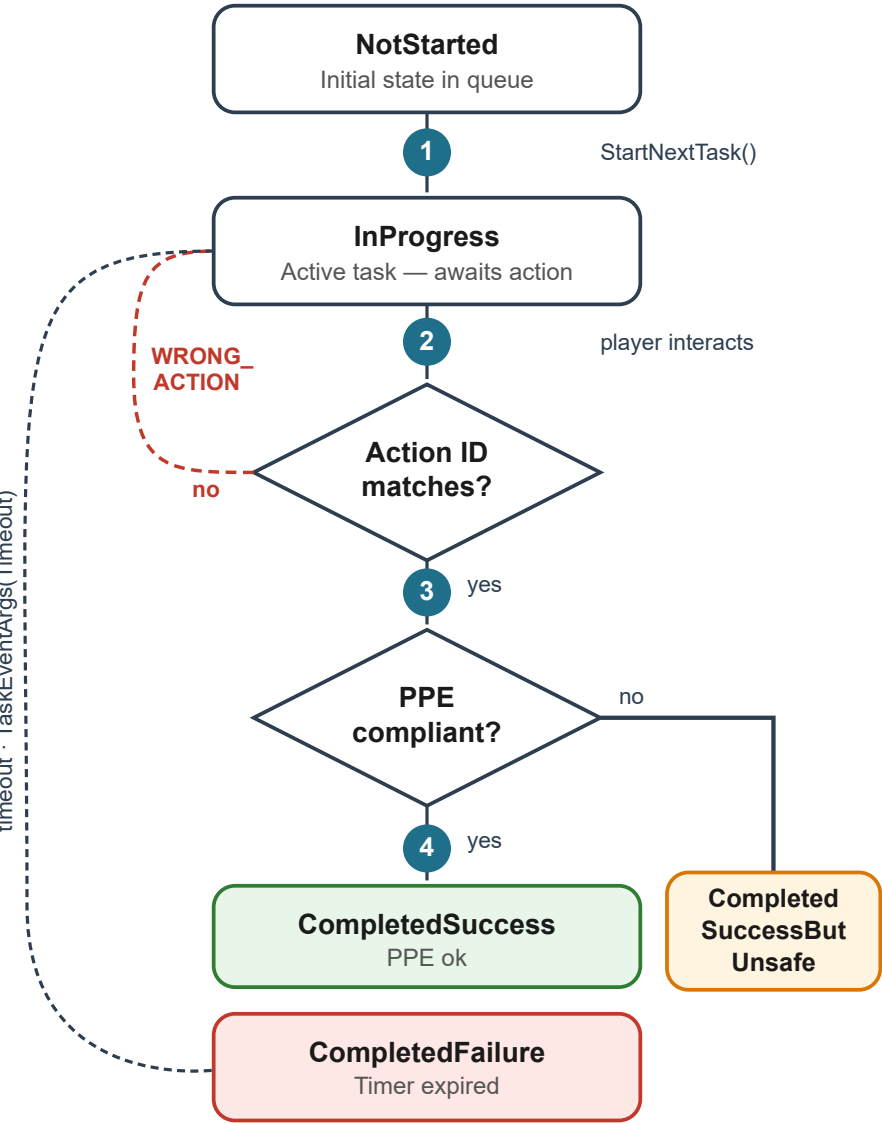


Task Lifecycle in the EventBus-Based Architecture

Sequential and Free-Order execution modes share a central publish/subscribe channel

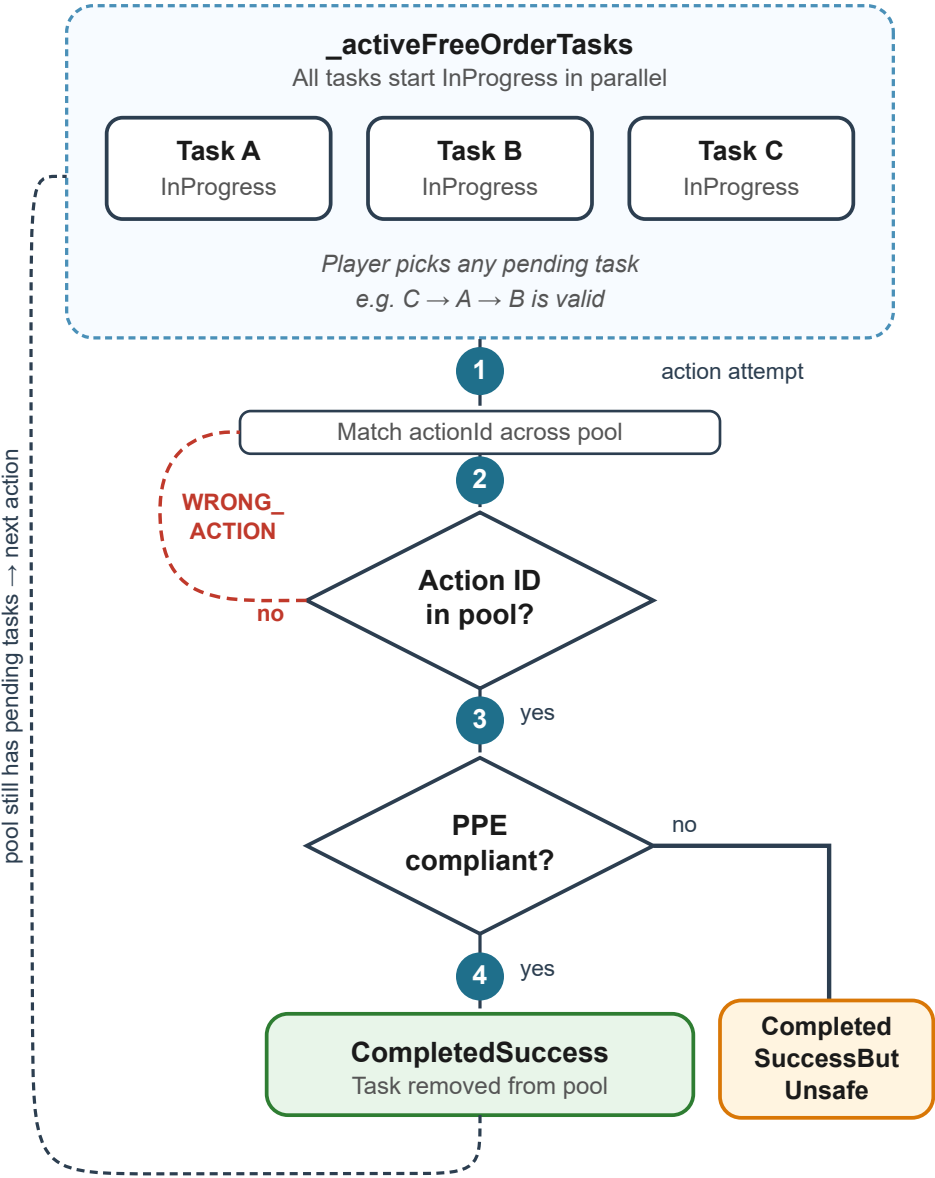
SEQUENTIAL MODE



Events emitted

- (1) **TaskEventArgs · phase = Started**
- (2) **ActionAttemptedEvent** (from ActionTrigger)
- (3) **TaskEventArgs · phase = Completed**
- SafetyViolationEventArgs · WRONG_ACTION / PPE_MISSING**

FREE-ORDER MODE



InspectionGateValidator (optional)

Triggered by RayInteractable "Begin Work" — verifies that every task in the group reached a terminal state.

Events emitted

- (1) **TaskGroupEventArgs · phase = Started** (per group)
- (2) **ActionAttemptedEvent** (consumed by engine)
- (3) **TaskEventArgs · phase = Completed** (per task)

Event Bus
Decoupled publish/subscribe channel — every state transition flows as a typed event
TaskEventArgs · ActionAttemptedEvent · PPEStateChangedEventArgs · TaskGroupEventArgs · SafetyViolationEventArgs

Components (subscribers/publishers on the bus)

TaskManagerCore
Owns RuntimeSafetyTask state
Publishes TaskEventArgs(Started)
Subscribes Completed/Timeout

ActionTrigger
VR interactable — fires canonical action IDs
Publishes ActionAttemptedEvent

SafetyRuleEngineCore
Validates actionId against current task / pool
Publishes Completed + violations

PPEManager
IPPEComplianceChecker queried during validation
PPEStateChangedEventArgs

InspectionGateValidator
Free-Order only — player-driven checkpoint, polls pending tasks

TimerSystem
Per-group time limit
Raises TaskEventArgs(Timeout)

ScoreService
Subscribes Completed/Violation
Updates running score

Evaluator HUD/WS
Mirrors all events to the dashboard via WebSocket

Legend

- State transition
- Event flow via EventBus
- SafetyViolation (loops back)
- Ordered step in canonical happy path
- Free-order pool (parallel tasks)
- Optional inspection gate

Sequential mode enforces a single `_activeSequentialTask`; Free-Order keeps all tasks of a group pending in `_activeFreeOrderTasks` and removes them on completion.